

Cross-Track Integration Evidence

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AnalystLab Africa — Experience Lab | HealthConnect Clinic Project | Data Analytics Track | Week 6

1. Track Collaborated With

Data Science.

2. Project Dependency

The Data Science track's Week 5 baseline no-show prediction model requires validated, high-signal features to prioritise during its Week 6 feature refinement and model improvement work.

3. Information/Output Received

The Data Science track's Week 5 problem definition: the model is a binary classifier predicting appointment outcome as No-Show vs. Attended. This shaped how the feature recommendations below were framed and prioritised.

4. Information/Output Provided

A validated, ranked feature recommendation, based on Week 6 advanced analysis, for use in the Data Science track's feature selection process:

- `previous_no_shows` (highest priority): Confirmed as the strongest, most consistent predictor of no-shows, holding true across all six patient age groups with no exceptions (43.5% to 68.8% no-show rate as prior no-shows increase overall).
- `distance_to_clinic_km` x `previous_no_shows` (recommended as an interaction feature, second priority): Week 6 analysis found distance strongly affects patients with 0-2 previous no-shows (rate climbing from ~42% to ~64% with distance), but has little additional effect on patients who already have 3+ previous no-shows, whose risk is already near-ceiling (89.5% at short distances). A single interaction feature is recommended over treating the two variables as independent.
- `appointment_type` (supporting feature): Diagnostic Test appointments showed a distinctly steeper distance sensitivity (39.4% to 61.7%) than other appointment types, suggesting this field adds useful signal.
- `reminder_sent` (lower priority): Only a modest standalone effect (4-point difference) was found in Week 5; unlikely to add strong predictive power alone.

- `waiting_time_minutes` (not recommended): Week 5 analysis found no meaningful relationship between waiting time and next-visit attendance; excluding this feature is recommended to avoid adding noise to the model.

5. Integration Activity Completed

Conducted segment-level validation of the Week 5 findings (by age group and appointment type) and combined-variable analysis (previous no-shows x distance) specifically to produce evidence-based, rather than assumed, feature guidance for the Data Science track's Week 6 model improvement work.

6. What Changed As a Result

The candidate feature list for the HealthConnect no-show prediction model moved from an untested, assumed list (Week 4/5) to a ranked, evidence-based list with a specific structural recommendation — testing a distance x prior-history interaction feature rather than including both variables independently. This directly informs how the Data Science track should approach feature engineering in its own Week 6 work.

7. Evidence

This document, together with Section 2 (Advanced Analysis) and Section 5 (Cross-Track Integration) of the Week 6 Track-Specific Output, constitutes the analytical evidence supporting this integration. The underlying calculations were performed on the HealthConnect Appointment Dataset, cross-validated across six age groups and four appointment types.